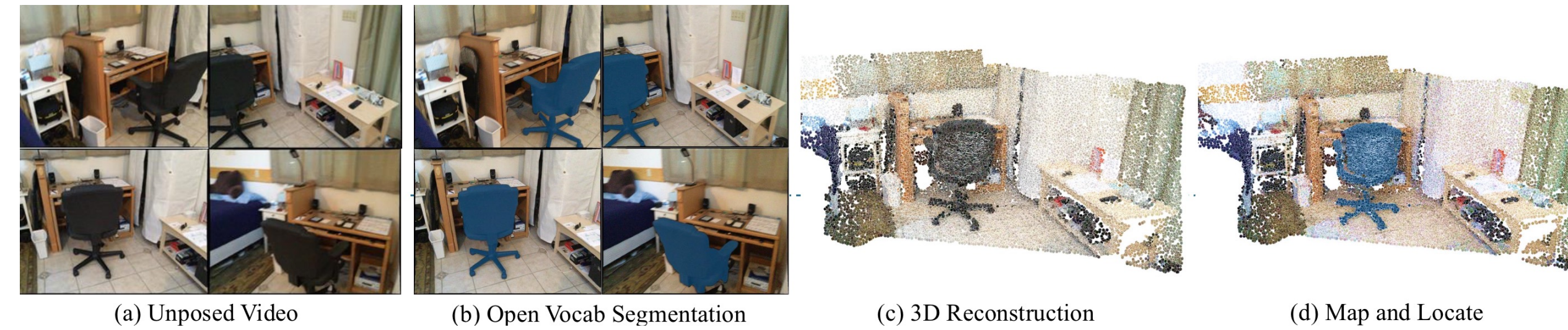




Map and Locate

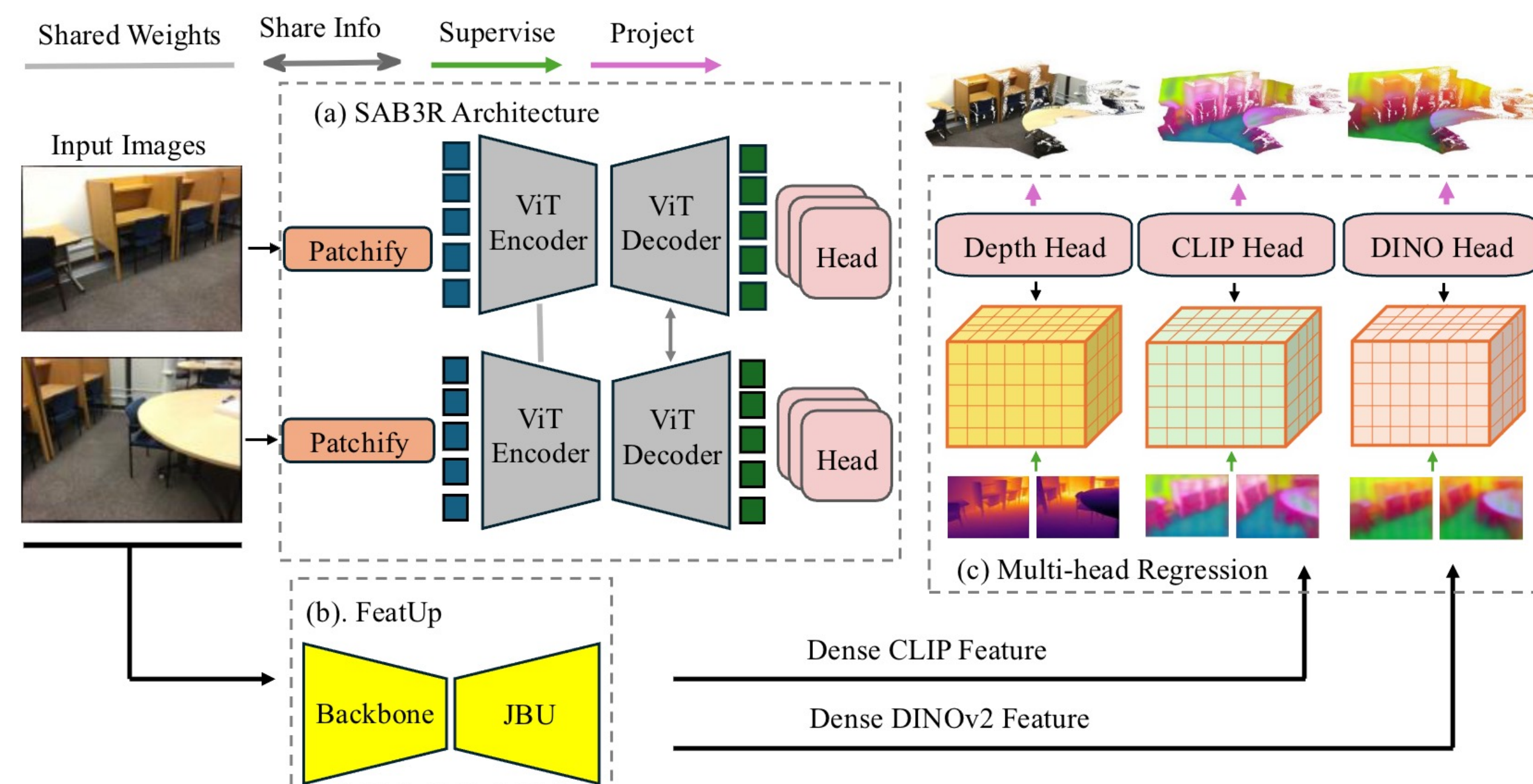


Map and Locate unifies the traditionally distinct objectives of

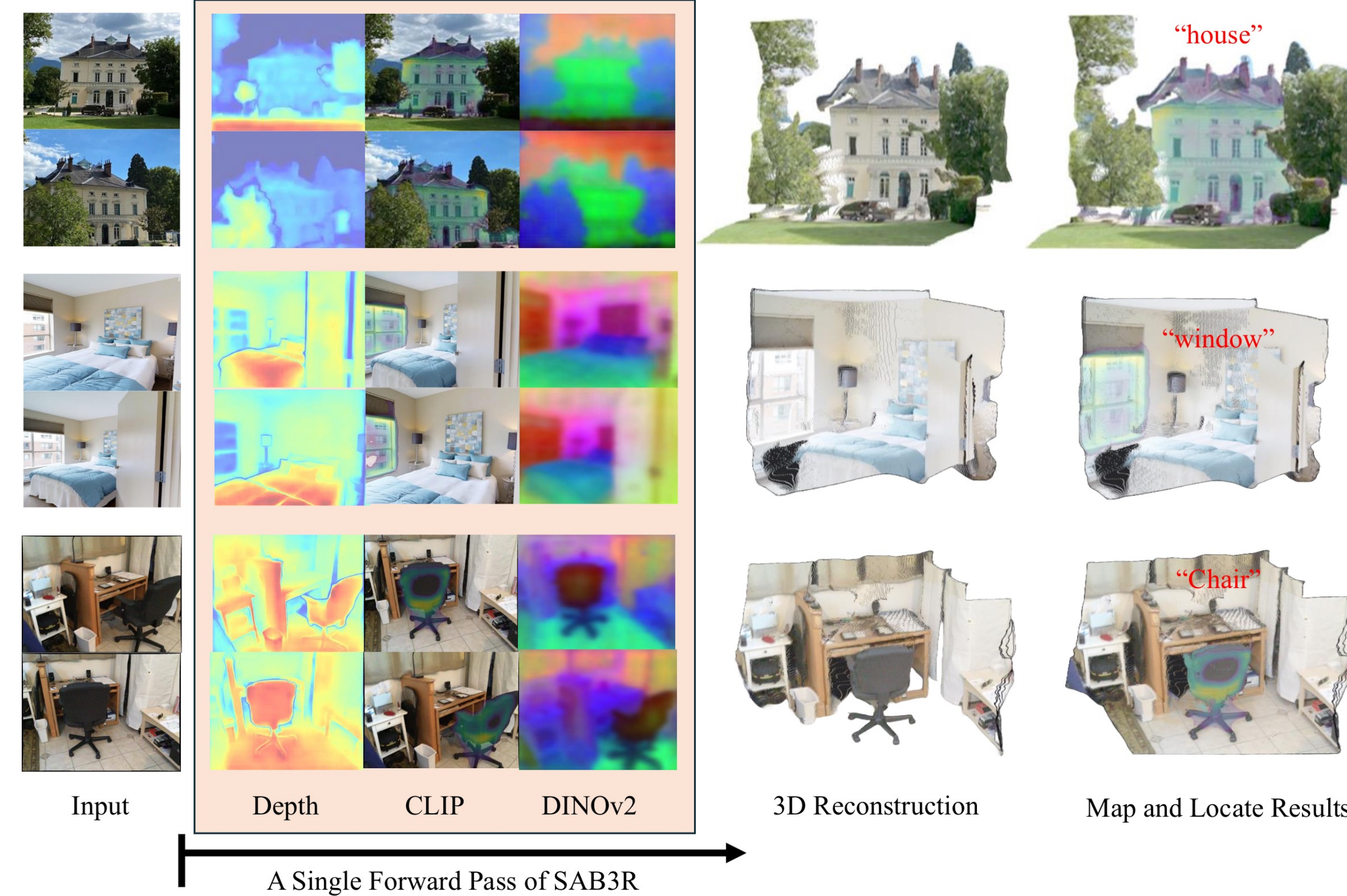
- open-vocabulary segmentation—detecting and segmenting object instances based on natural language queries
- 3D reconstruction, the process of estimating a scene's 3D structure from visual inputs.

This task serves as a critical step toward real-world embodied AI applications and introduces a practical task that bridges reconstruction, recognition and reorganization.

Methods Architecture



SAB3R Pipeline



Our method, *SAB3R*, a semantic-augmented backbone for 3D reconstruction, enables zero-shot open-vocabulary segmentation and 3D reconstruction from unposed images in a single forward pass. By jointly performing reconstruction and open-vocabulary semantic segmentation, *SAB3R* introduces a novel capability that unifies these tasks within a single framework. *SAB3R* simultaneously generates depth, dense DINOv2, and dense CLIP features, which are then used for *Map and Locate* task.

Experiments on Map and Locate

Model	Params	FLOPs	Sparse View = 2				Sparse View = 3				Sparse View = 4			
			mIoU	Acc.	mComp.	mdComp.	mIoU	Acc.	mComp.	mdComp.	mIoU	Acc.	mComp.	mdComp.
Baseline	838M	248G	4.57	18.10	0.64	0.67	6.03	21.26	0.68	0.71	5.12	19.31	0.68	0.70
LSM [16]	1B	> 592G	21.40	42.34	0.72	0.80	-	-	-	-	-	-	-	-
SAB3R (C)	729M	218G	17.26	41.11	0.73	0.75	22.83	53.19	0.78	0.81	19.92	48.07	0.77	0.80
SAB3R (CD)	729M	218G	17.50	42.72	0.73	0.76	22.94	52.86	0.77	0.80	20.31	46.26	0.75	0.78

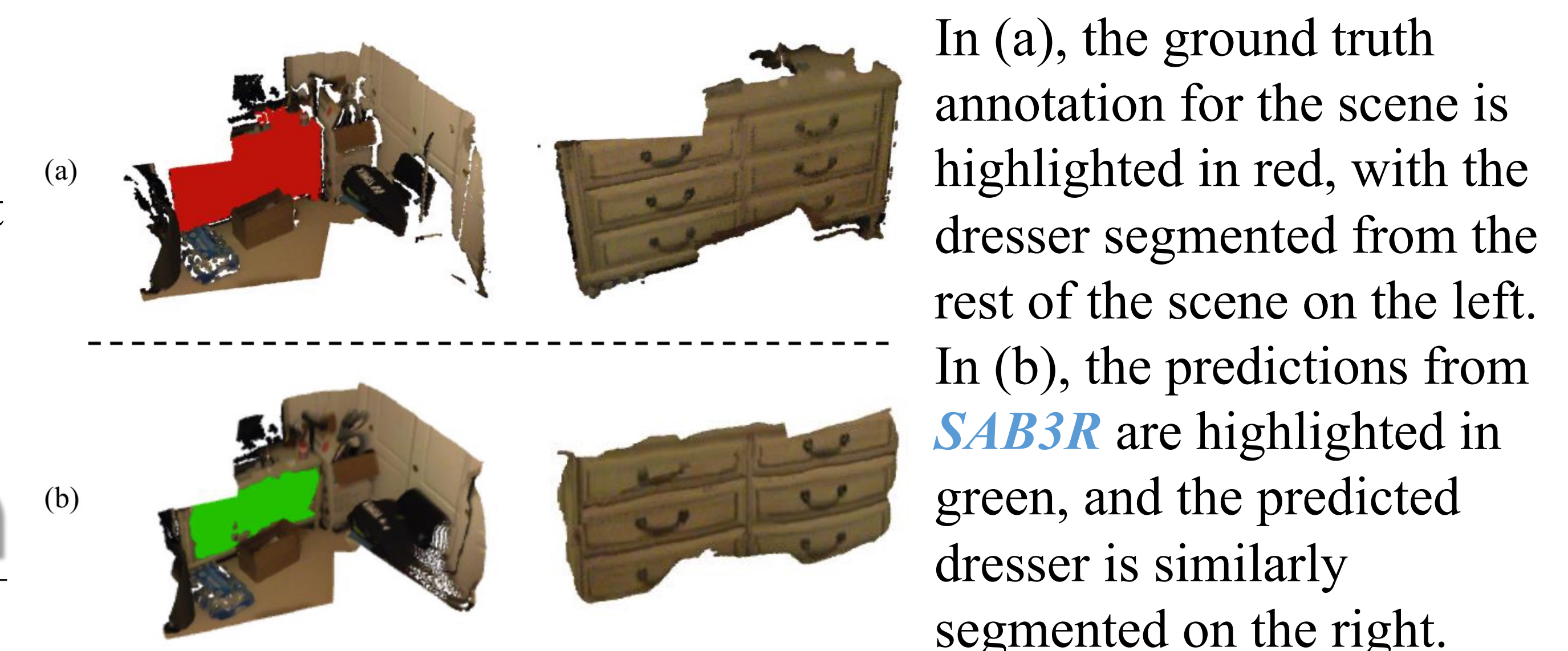
Additional Experiments

Model	Arch	VOC↑	ADE20k↑
GroupViT [66]	ViT-S	52.3	-
ViewCo [47]	ViT-S	52.4	-
ViL-Seg [34]	ViT-B	37.3	-
OVS [67]	ViT-B	53.8	-
CLIPpy [44]	ViT-B	52.2	13.5
TCL [8]	ViT-B	51.2	14.9
SegCLIP [35]	ViT-B	52.6	8.7
SAM-CLIP [61]	ViT-B	60.6	17.1
FeatUp (MaskCLIP)	-	51.2	14.3
SAB3R (C)	ViT-B	55.4	18.3
SAB3R (CD)	ViT-B	56.4	19.0

Methods	Train	NYUD-v2 (Indoor)		KITTI (Outdoor)	
		Rel↓	$\delta_{1.25} \uparrow$	Rel↓	$\delta_{1.25} \uparrow$
DPT-BEiT[45]	D	5.40	96.54	9.45	89.27
NeWCRFs[72]	D	6.22	95.58	5.43	91.54
Monodepth2 [20]	SS	16.19	74.50	11.42	86.90
SC-SfM-Learners [5]	SS	13.79	79.57	11.83	86.61
SC-DepthV3 [57]	SS	12.34	84.80	11.79	86.39
MonoViT [77]	SS	-	-	9.92	90.01
RobustMIX [40]	T	11.77	90.45	18.25	76.95
SlowTv [55]	T	11.59	87.23	(6.84)	(56.17)
DUST3R 224-NoCroCo	T	14.51	81.06	20.10	71.21
DUST3R 224	T	10.28	88.92	16.97	77.89
DUST3R 512	T	6.51	94.09	12.02	83.43
MAS3R	T	8.17	92.59	8.28	93.27
SAB3R (C)	T	7.80	92.67	11.63	86.74
SAB3R (CD)	T	7.67	92.82	12.53	83.51

SAB3R demonstrates strong performance across zero-shot semantic segmentation, monocular depth estimation, and multi-view pose regression—on par with, and slightly behind our teacher models.

Qualitative Example



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